

**Building and Breaking Classical Ciphers: A Comparison of Caesar and Substitution
Encryption**

Ruthvik Nath Bandari

Northeastern University

AAI6680: AI for Cybersecurity

Mimoza Dimodugno, PhD

September 27, 2026

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Encryption

Encryption converts readable plaintext into ciphertext so that only holders of the key can recover the message. This lab implemented two classical ciphers, the Caesar shift and the simple substitution cipher, and then attacked both. The goal was to measure *why* one cipher is stronger than the other, and why neither is secure.

Method

Data and Preprocessing

The attacks compare ciphertext against a statistical model of English, so two inputs were loaded and validated first. The first was Lewand's (2000) letter-frequency table; the code checks that it covers all 26 letters and sums to 100%. The second was an original five-paragraph corpus of 2,304 letters. For analysis only, text was normalized to uppercase A–Z. The ciphers themselves preserve case, spaces, digits, and punctuation, and they reject invalid input such as missing values, non-string types, and substitution keys that are not permutations.

Cipher Implementation

The Caesar cipher shifts each letter by a key $k \pmod{26}$. The substitution cipher maps A–Z through a randomly shuffled alphabet generated from a fixed seed, and decryption applies the inverse mapping. Assertions verify every required test case, including “Hello World” → “Khoor Zruog”.

Attack Logic

Each candidate decryption was scored with Pearson's chi-squared statistic against English letter frequencies, and the lowest score was selected. I chose chi-squared over the common “most frequent letter is E” heuristic because it uses all 26 letters and heavily penalizes rare letters (Q, X, Z) appearing too often, which is exactly what an incorrect shift produces. For the substitution cipher, the attack combined frequency rank-matching, common-word patterns (*THE*, *AND*), and a known-word “crib” located by its letter pattern.

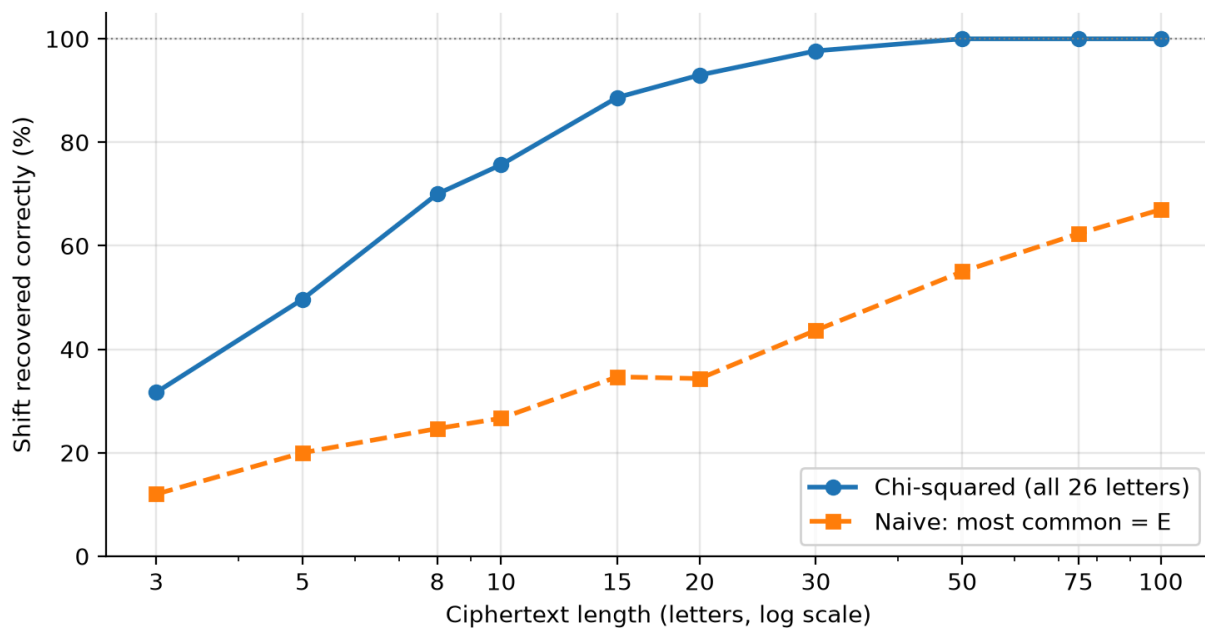
Results

Both ciphers encrypted and decrypted all test cases correctly. Run time scaled linearly with message length, and substitution was about one-third faster on long inputs (0.07 vs. 0.11 μ s per character), so speed does not distinguish the two methods.

The automated Caesar breaker recovered a hidden shift from a single sentence: the correct candidate scored $\chi^2 = 14.9$, compared with 249.5 for the runner-up. Across 300 random trials per message length (see Figure 1), chi-squared recovered the key 93% of the time at 20 letters and 100% of the time from 50 letters onward. The naive E-rule achieved only 34% and 55% at those lengths.

Figure 1

Caesar Breaker Accuracy by Ciphertext Length

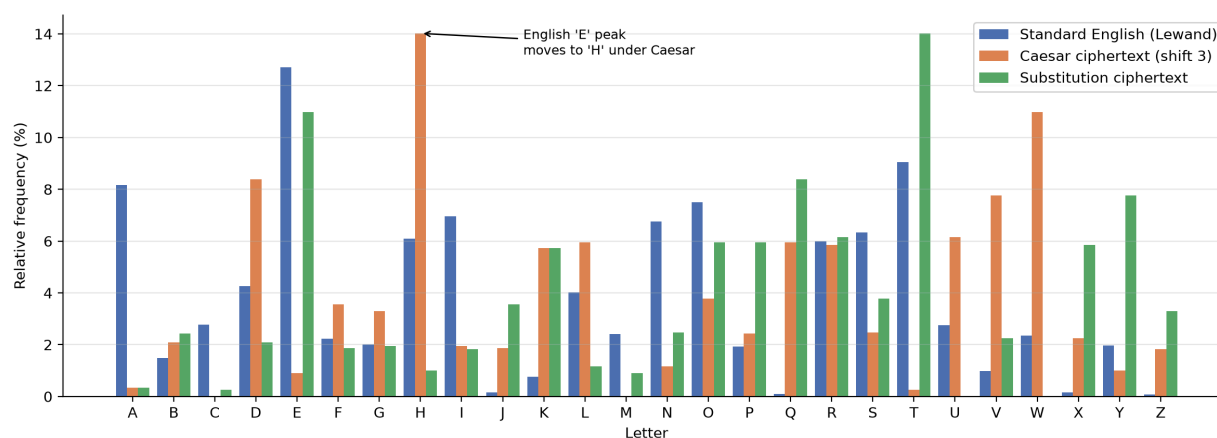


Note. Each point is the share of 300 random corpus excerpts whose Caesar shift was recovered correctly. The solid line uses chi-squared scoring across all 26 letters; the dashed line assumes the most frequent ciphertext letter is E. Corresponds to Figure 2 in the accompanying notebook.

Against the substitution cipher, frequency ranking alone decrypted 42.8% of ciphertext letters correctly (see Figure 2). Assuming that the two most common three-letter words were *THE* and *AND* raised this to 51.8%. A single guessed word, “encryption,” matched exactly one ciphertext word by shape and raised accuracy to 89.4%, which left the text readable.

Figure 2

Letter Frequency in Standard English Compared With Caesar and Substitution Ciphertext



Note. English frequencies are from Lewand (2000). Both ciphertexts encrypt the same five-paragraph corpus; the Caesar key is a shift of 3. Corresponds to Figure 3 in the accompanying notebook.

Security Analysis

Table 1 summarizes the comparison. Substitution’s key-space advantage of roughly 84 bits is irrelevant in practice, because the attacker never needs to search the keys. Figure 2 shows why: Caesar slides the English histogram sideways and substitution shuffles it, but once sorted, both ciphertext distributions are identical to the plaintext distribution. Repeated words also encrypt identically, and letter patterns such as the double *s* in *password* survive, so a single guessed word can be located by its shape alone.

Table 1*Security Comparison of the Caesar and Substitution Ciphers*

Criterion	Caesar cipher	Substitution cipher
Key space	25 keys (≈ 4.6 bits)	$26! \approx 4.0 \times 10^{26}$ keys (≈ 88 bits)
Brute force at 10^9 keys/s	Nanoseconds	$\approx 1.3 \times 10^{10}$ years
Known plaintext	One letter reveals the key	One word reveals only its letters (36% here)
Repeated words and patterns	Leaked	Leaked
Frequency analysis	Broken instantly	Broken with one paragraph

Note. Brute-force times assume one billion key trials per second. The known-plaintext figure is the share of message letters revealed by the single known word “Friday.”

Discussion

Caesar falls to brute force in a fraction of a second. Substitution survives brute force but still falls to statistical analysis, because both ciphers are deterministic and monoalphabetic: they relabel letters without hiding their frequencies or structure. The key lesson is that a large key space is necessary but not sufficient. Secure modern ciphers such as AES-GCM also require diffusion and randomization, so that ciphertext is statistically indistinguishable from random noise (Stallings, 2017).

The breaker is itself a small statistical classifier that scores inputs against a model of “normal,” the same principle that underlies the anomaly-based intrusion detector from Module 1. A limitation of this lab is that all results come from a single English corpus; an automated hill-climbing substitution solver would be the natural extension.

References

Lewand, R. E. (2000). *Cryptological mathematics*. Mathematical Association of America.

Stallings, W. (2017). *Cryptography and network security: Principles and practice* (7th ed.).
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